

Assessment-1

Testing on Live Application-

Section A: Scenario Based Conceptual Questions

Q1. An online learning platform releases new courses every month. Identify the most suitable Software Development Life Cycle (SDLC) model and justify your choice.

Answer: The **Agile SDLC Model** is the most suitable model for an online learning platform.

Justification:

- New courses are added every month, so frequent updates are required.
- Agile allows development in small iterations called sprints.
- Customer feedback can be collected after every release.
- New features and improvements can be added quickly.
- Bugs can be fixed faster without waiting for the entire project to finish.

Therefore, Agile is the best choice because it supports continuous development and regular releases.

Q2. For a course purchase feature, define testing objectives from a Quality Assurance perspective.

Answer: The testing objectives for the Course Purchase feature are:

- Verify that users can purchase a course successfully.
- Verify that the correct course price is displayed.
- Verify that discounts and coupon codes work correctly.
- Verify that payment is processed successfully.
- Verify that users receive a purchase confirmation.
- Verify that purchased courses appear in "My Courses."
- Verify that payment failures are handled properly with appropriate error messages.
- Verify that user data and payment information remain secure.

Q3. Explain how Non-Functional Testing contributes to a better learner experience.

Answer: Non-Functional Testing checks how well the application performs instead of what it does.

It helps improve the learner experience by:

- ✓ Ensuring the website loads quickly.
- ✓ Checking that videos play smoothly without buffering.
- ✓ Making sure the application works well with many users at the same time.
- ✓ Ensuring the application is secure and protects user information.
- ✓ Verifying that the application works on different devices and browsers.

Good non-functional testing improves performance, security, reliability, and user satisfaction.

Q4. During sprint execution, how should testers ensure acceptance criteria are testable?

Answer: Testers should review the acceptance criteria at the beginning of the sprint with the Product Owner and developers.

They should:

Make sure every requirement is clear and understandable.

Remove any ambiguity from the requirements.

Create test cases based on the acceptance criteria.

Discuss missing scenarios with the development team.

Confirm that each acceptance criterion can be verified through testing.

This helps reduce misunderstandings and ensures the feature meets business requirements.

Q5. A payment failure occurs after deployment. Assign Severity and Priority with justification.

Answer:

Severity: Critical

Priority: High

Justification:

A payment failure directly affects the main functionality of the application because users cannot purchase courses successfully. It can lead to customer complaints, financial loss, and reduced trust in the platform.

Since it impacts many users and affects business revenue, it should be fixed immediately. Therefore, the severity is **Critical** and the priority is **High**.

Section B: Practical Manual Testing Tasks

Q1. STLC (Student Login): Document the full lifecycle including Requirement Analysis, Planning, Test Design, Execution, and Closure. Define specific Entry and Exit criteria for the Student Login module.

Answer: Software Testing Life Cycle (STLC) for Student Login Module

1. Requirement Analysis

- Study the Student Login requirements.
- Understand the login functionality.

2. Test Planning

- Prepare the Test Plan.
- Define testing scope and objectives.
- Identify the test environment and test data.

3. Test Case Design

- Create test scenarios.
- Write positive and negative test cases.

4. Test Execution

- Execute all test cases.
- Compare Actual Result with Expected Result.
- Log defects if any issue is found.

5. Test Closure

- Verify all defects are fixed.
- Prepare Test Summary Report.
- Close the testing process after successful verification.

Entry Criteria

- Requirements are finalized.
- Login module is developed.
- Test environment is ready.
- Test data is available.

Exit Criteria

- All test cases are executed.
- Critical and Major defects are fixed.
- Test Summary Report is prepared.

- Login module is approved for release.

Q2. Test Design (Course Rating 1–5): Apply Equivalence Partitioning and Boundary Value Analysis to the 1–5 star rating input. List the specific Valid and Invalid numeric test data required to verify the input limits.

Answer:

Equivalence Partitioning (EP)

Valid Test Data

1

2

3

4

5

Invalid Test Data

0

-1

6

7

10

Boundary Value Analysis (BVA)

Test Data Status

0	Invalid
1	Valid (Minimum Boundary)
2	Valid
4	Valid
5	Valid (Maximum Boundary)
6	Invalid

Q3. Test Cases (Course Experience): Write one functional test case for each: Course Search & Browse, Course Learning Experience, and Assessments & Quizzes. Each case must include Steps, Expected Result, and Pass/Fail Status.

Answer:

Test Case 1 – Course Search & Browse

Test Case ID: TC_001

Module: Course Search & Browse

Objective: Verify that the user can search for a course successfully.

Steps:

1. Open the Online Learning Platform.
2. Enter "Software Testing" in the search box.
3. Click the Search button.

Expected Result: The matching course should be displayed in the search results.

Actual Result: As expected

Status: Pass

Test Case 2 – Course Learning Experience

Test Case ID: TC_002

Module: Course Learning Experience

Objective: Verify that the user can access and watch the course videos.

Steps:

1. Login with valid credentials.
2. Open "My Courses."
3. Select an enrolled course.
4. Play the course video.

Expected Result: The video should play successfully without any interruption.

Actual Result: As expected

Status: Pass

Test Case 3 – Assessments & Quizzes

Test Case ID: TC_003

Module: Assessments & Quizzes

Objective: Verify that the user can submit a quiz successfully.

Steps:

- 1) Login to the application.
- 2) Open the enrolled course.
- 3) Start the quiz.
- 4) Answer all questions.
- 5) Click the Submit button.

Expected Result: The quiz should be submitted successfully and the score should be displayed.

Actual Result: As expected

Status: Pass

Q4. Defect Log: Document 3 unique defects found in the modules above. Each report must include: Defect Title, Severity, Steps to Reproduce, Actual vs Expected Result, and Environment details.

Answer:

Defect 1

Defect Title: Login button remains enabled when Email and Password fields are empty

Severity: Major

Priority: Medium

Environment: MacOS, Google Chrome Version 137

Steps to Reproduce:

1. Open Student Login page.
2. Leave Email and Password fields empty.
3. Observe the Login button.

Expected Result:

Login button should remain disabled until valid credentials are entered.

Actual Result:

Login button is enabled even when both fields are empty.

Defect 2

Defect Title: Course search does not display available course

Severity: Major

Priority: High

Environment: Android 16, Chrome Browser

Steps to Reproduce:

- 1) Login to the application.
- 2) Search for "Software Testing."
- 3) Click Search.

Expected Result:

The Software Testing course should appear in the search results.

Actual Result:

"No Course Found" message is displayed even though the course exists.

Defect 3

Defect Title: Quiz score is not displayed after successful submission

Severity: Critical

Priority: High

Environment: MacOS 16, Microsoft Edge Browser

Steps to Reproduce:

1. Login with valid credentials.
2. Open an enrolled course.
3. Start the quiz.
4. Answer all questions.
5. Submit the quiz.

Expected Result:

The quiz score should be displayed immediately after submission.

Actual Result:

The quiz is submitted successfully, but the score is not displayed.

Section C: Mini Capstone Project – End-to-End Testing Case

Q1. Test Planning Document: Prepare a Test Plan covering scope, objectives, testing types, environment, risks, assumptions, and entry and exit criteria.

Answer:**Test Plan****Project Name**

Online Learning Platform

Scope

Testing the Student Login, Course Selection, Payment, and Course Access modules.

Objectives

- i. Verify that all features work correctly.
- ii. Ensure users can purchase and access courses successfully.
- iii. Identify and report defects before release.
- iv. Improve software quality.

Testing Types

- 1) Functional Testing
- 2) UI Testing
- 3) Integration Testing
- 4) System Testing
- 5) Regression Testing
- 6) Smoke Testing

Test Environment

- Operating System: MacOS / Android 14
- Browser: Google Chrome
- Internet: Stable Wi-Fi Connection
- Database: Test Database

Risks

- ◆ Delay in development.
- ◆ Server downtime.
- ◆ Payment gateway issues.
- ◆ Requirement changes during testing.

Assumptions

- ✓ Application is deployed in the test environment.
- ✓ Test data is available.
- ✓ All required modules are completed.
- ✓ Payment gateway sandbox is available.

Entry Criteria

- ✓ Requirements are approved.
- ✓ Build is deployed.
- ✓ Test environment is ready.
- ✓ Test cases are prepared.

Exit Criteria

- ✓ All planned test cases are executed.
- ✓ Critical defects are fixed.
- ✓ Test Summary Report is prepared.
- ✓ Application is approved for release.

Q2. Requirements Traceability Matrix: Create a Requirements Traceability Matrix mapping requirements to test scenarios, test cases, and defects.

Answer:

Requirement ID	Requirement	Test Scenario	Test ID	Case ID	Defect ID
R001	Student Login	Verify successful login	TC001		D001
R002	Course Search	Verify course search	TC002		D002
R003	Course Purchase	Verify successful payment	TC003		D003
R004	Course Access	Verify purchased course is accessible	TC004		D004

Q3. End-to-End Test Scenario: Write one detailed End-to-End test scenario covering Login → Course Selection → Payment → Course Access.

Answer:

Test Scenario ID: TS001

Scenario Name

Verify that a student can successfully purchase and access a course.

Precondition

- Student account is registered.
- Internet connection is available.

Test Steps

- 1) Open the Online Learning Platform.
- 2) Login using valid Email and Password.
- 3) Search for the "Software Testing" course.
- 4) Open the course details page.
- 5) Click **Buy Now**.
- 6) Select UPI as the payment method.
- 7) Complete the payment successfully.
- 8) Open **My Courses**.
- 9) Select the purchased course.
- 10) Play the first video lesson.

Expected Result

- ✓ Login is successful.
- ✓ Course is displayed.
- ✓ Payment is completed successfully.
- ✓ Purchased course appears in **My Courses**.
- ✓ Student can watch the course without any issue.

Actual Result: All steps are executed successfully.

Status: Pass

Q4. Test Summary Report: Prepare a Test Summary Report including execution status, defects identified, and release recommendation.

Answer:

Test Summary Report

Project Name

Online Learning Platform

Testing Period

_____ to _____

Total Test Cases: 10

Executed Test Cases: 10

Passed Test Cases: 9

Failed Test Cases: 1

Blocked Test Cases: 0

Defects Identified

- ✧ Major Defects: 1
- ✧ Minor Defects: 2
- ✧ Critical Defects: 0

Overall Test Status

Most of the application features are working correctly. The identified defects have been reported to the development team.

Release Recommendation

The application is **recommended for release after fixing the reported Major defect and completing regression testing.**

Prepared By--

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